The effect of vegetarian diet on selected essential nutrients in children.

INTRODUCTION: Vegetarian diets are considered to promote health and reduce the risk of some chronic diseases. It is also known that restriction or exclusion of animal foods may result in low intake of essential nutrients. The aim of the presented study was to assess the intake and serum status of vitamin B12, folate, vitamins A, E and D, as well as concentrations of homocysteine, total antioxidant status and iron balance in Polish vegetarian children. MATERIAL AND METHODS: The study included 50 children, aged 5-11 who had been referred to the Institute of Mother and Child for dietary consultation. From those, 32 were vegetarians (aged 6.5±4.2 years) and 18 omnivores (aged 7.9±2.7 years). Dietary constituents were analyzed using the nutritional programme Dietetyk2®. Folate and vitamin B12 were determined with a chemiluminescence immunoassay, total homocysteine with a fluorescence polarization immunoassay and TAS (total antioxidant status) by colorimetric method. Vitamin A and E in serum were determined by the high-pressure liquid chromatography method (HPLC) and vitamin D by immunoenzymatic assay (ELISA). Concentrations of iron, ferritin, transferrin and total iron-binding capacity (TIBC) in serum were determined by commercially available kits. RESULTS: In vegetarian children daily intake of vitamin B12 (1.6 ■g) was in the recommended range, that of folate (195 ■g) and vitamin A (1245 ■g) higher, but vitamin E slightly lower (6.6 ■g) and three-fold lower vitamin D (1.1 ■g) than references allowance. Serum concentrations of vitamin B12 (548 pg/ml), folate (12.8 ng/ml), vitamin A (1.2 ■mol/L), vitamin E (15.6 ■mol/l) were within physiological range, but that of vitamin D (13.7 ■g/L) was only half of the lowest limit of the reference value. In vegetarian children in comparison to omnivorous similar levels of homocysteine (6.13 ■mol/L vs 5.45 ■mol/L) and vitamin A (1,17 ■mol/L vs 1.32 ■mol/L) were observed. Lower ($p<0.05$) values of vitamin E (15.6 ■mol/L vs 18.4 ■mol/L) and TAS (1.21 mmol/L vs 1.30 mmol/L; $p<0.0001$) were found. Concentrations of iron markers were in physiological range. CONCLUSION: Obtained results indicated that intakes of vitamin B12 and folic acid from vegetarian diets are sufficient to maintain serum concentrations of both homocysteine and iron in the range observed in omnivorous children. High consumption of vitamin A and low vitamin E only slightly affected their serum values. Significantly lower concentration of serum vitamin E in vegetarian children in comparison to nonvegetarians may be reflected with statistically significant lowering of total antioxidant status. Insufficient intake of vitamin D and its low serum concentration should be under close monitoring in vegetarian children. In order to prevent vitamin D deficiency appropriate age-dependent supplementation should be considered.